



NASA confirms lunar water

NASA has confirmed that the sunlit side of the lunar surface does hold water, and ice in small pockets. <http://digit.in/nov20-27>



Tagging an asteroid

NASA has shared images showing their OSIRIS-REx spacecraft make contact with an asteroid. <http://digit.in/nov20-28>

The PSLV-C37 taking off with 104 satellites on board

The Small Satellite Launch Vehicle

ISRO is not only developing a new launch vehicle to serve an emerging market for micro and nano satellite launches, but is also opening up a dedicated space port for the new rocket.

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Over the course of the last five years, India has launched over 250 “customer” satellites, which are contracted by two private companies that have spun out of the Indian Space Research Organisation (ISRO), the older Antrix Corporation, and the newer New Space India Limited (NSIL) which bagged its first contract in 2019. Typically, these satellites are deployed into space as secondary payloads on board the Polar Satellite Launch Vehicle (PSLV) missions, which is considered to be the workhorse rocket of the Indian space agency. ISRO offers competitively priced launched services, and the per kg launch prices for smaller satellites are actually lesser than say

SpaceX, a cutting edge private spaceflight agency that reuses its rockets to bring down the cost of every launch. ISRO has never failed to deploy a contracted satellite. The PSLV has enjoyed a close to perfect success rate, with only the PSLV-C39 mission failing to deploy a navigation satellite, part of the NAVIC constellation, which would provide local navigation capabilities to India. Note that this satellite was one of ISRO's own, and not a contracted payload.

ISRO launches a lot of satellites, and Australia, Canada, Finland, France, Germany, Italy, Japan, Israel, Korea, Lithuania, Switzerland and the UK are all what you might call repeat customers. ISRO has also launched satellites for the Netherlands, Colombia, Kazakhstan, Latvia, the Czech Republic, Chile and Indonesia.

Over the past five years, ISRO has deployed more than 200 American

satellites, even though American companies are not allowed to directly use launch services offered by India because the rates offered by Antrix and NSIL are so competitive, that the local launch services simply cannot offer better prices. The contracts are relayed through international intermediaries, such as Innovative Solutions, which first bag the contracts from American companies, and then hire ISRO for the actual launch. In fact, American companies can even use Indian launch services to deploy satellites that have not been cleared for launch by the US Federal Communications Commission, the regulatory body for satellite launches. This is exactly what a Silicon Valley startup called Swarm Technologies did when it sent up four SpaceBEE satellites with the PSLV C40 launch in early 2018. Two companies, Spire Global and Planet Labs keep launching satellites at every opportunity available, and are in the process of building up massive swarms that work together. Planet Labs has launched 152 Dove satellites, with 88 of them deployed during the record setting PSLV C37 flight in 2017. Spire